

Solve a quadratic equation

$$3\left(x^2 - \frac{2}{3}\right) = 4$$

↓ expand

$$3x^2 - 2 = 4$$

↓ +2

$$3x^2 = 6$$

↓ ÷3

$$x^2 = 2$$

↓ $\sqrt{\dots}$

$$\sqrt{x^2} = \sqrt{2}$$

↓ $\sqrt{x^2} = |x|$

$$|x| = \sqrt{2}$$

↓ so that

$$x = \pm\sqrt{2}$$

Isolate the term
with the variable.

Complete the square

$$y = 2x^2 - 3x + 5$$

Form a binomial square:
add and subtract $\frac{9}{16}$ inside.

$$y = 2\left(\underbrace{x^2 - \frac{3}{2}x + \left(\frac{3}{4}\right)^2}_{(a-b)^2 = a^2 - 2ab + b^2} - \underbrace{\left(\frac{3}{4}\right)^2 + \frac{5}{2}}_{\text{simplify}}\right)$$

$(a-b)^2 = a^2 - 2ab + b^2$

$$y = 2\left(\left(x - \frac{3}{4}\right)^2 + \frac{31}{16}\right)$$

simplify

$$y = 2\left(x - \frac{3}{4}\right)^2 + \frac{31}{8}$$

Real-valued x ; arrows indicate equivalent algebraic steps.